LEX PROGRAM TO FIND THE FREQUENCY OF EACH WORD

1. AIM

To write and execute a Lex program to count the frequency of each word in a given input text.

2. PSEUDOCODE

1. Start the program.
2. Initialize an empty list of words and their frequency counts.
3. Read the input text using the Lex scanner.
4. Whenever a word is matched, convert it to lowercase.
5. Search the list to check whether the word has already been stored.
6. If the word exists, increment its count; otherwise, add it with a count of 1.
7. When the end of input is reached, display each unique word and its frequency.
8. Stop.

3. LEX PROGRAM

```
%option noyywrap

%{
#include <stdio.h>
#include <string.h>
#include <ctype.h>

#define MAX_WORDS 1000
#define MAX_LEN 100

char words[MAX_WORDS][MAX_LEN];
int counts[MAX_WORDS];
int total = 0;
int i, found;
```

```
%}

%%
[A-Za-z]+ {
    char word[MAX_LEN];
    int j;

    strncpy(word, yytext, MAX_LEN - 1);
    word[MAX_LEN - 1] = '\0';

    for (j = 0; word[j]; j++)
        word[j] = tolower((unsigned char)word[j]);

    found = 0;

    for (i = 0; i < total; i++) {
        if (strcmp(words[i], word) == 0) {
            counts[i]++;
            found = 1;
            break;
        }
    }

    if (!found && total < MAX_WORDS) {
        strcpy(words[total], word);
        counts[total] = 1;
        total++;
    }
}

[ \t\n]+ ;
. ;
%%

int main(void) {
```

```
    printf("Enter text (Ctrl+D to finish):\n");
    yylex();

    printf("\nWord frequencies:\n");
    for (i = 0; i < total; i++)
        printf("%s: %d\n", words[i], counts[i]);

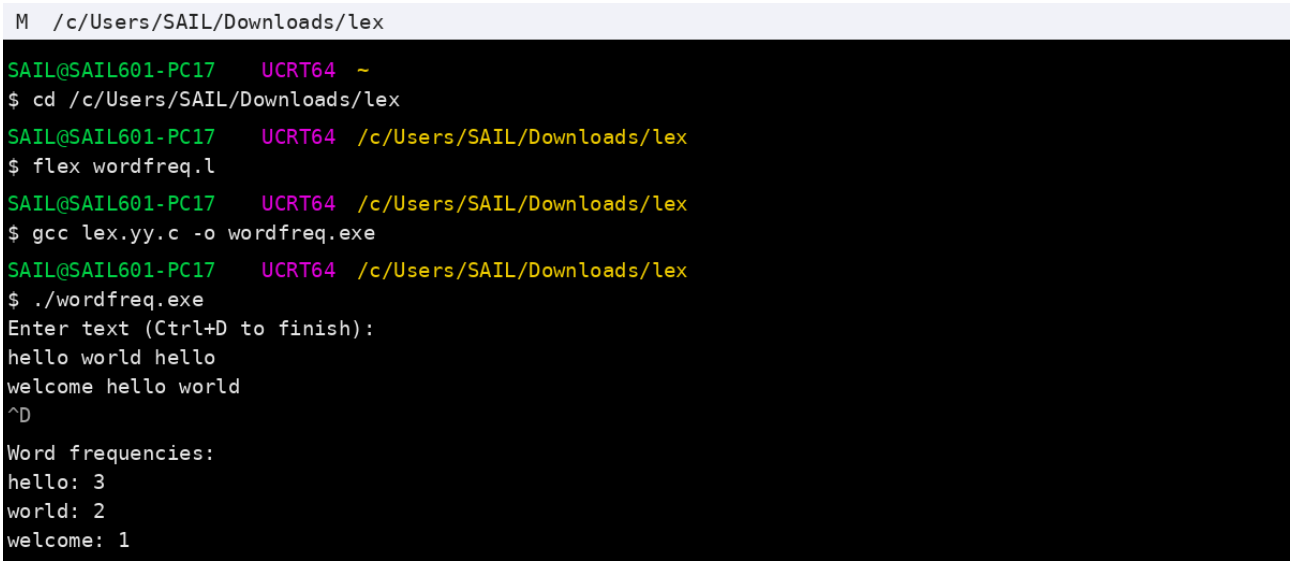
    return 0;
}
```

4. SAMPLE INPUT

hello world hello
welcome hello world

5. OUTPUT IMAGE

The following is an illustrative expected-output image for the sample input.



6. RESULT

Thus, the Lex program to find the frequency of each word in a given input text was written. For the sample input, the expected frequencies are: hello = 3, world = 2, and welcome = 1.